Gianluca Sabatini

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Education

ETH Zürich *September 2024 – Present*
MSc in Robotics, Systems and Control - Specialized MSc of Computer Science Department *Zurich, Switzerland*

- GPA: 5.6/6.0
- Focus on Machine Learning and Control Theory
- Coursework: Probabilistic AI, Machine Perception, LLMs, Computer Vision, Game Theory, MPC, Recursive Estimation

Alma Mater Studiorum - University of Bologna *September 2021 – July 2024*
BSc in Computer Engineering *Bologna, Italy*

- Final GPA: 29.3/30.0 - Final grade: 110 and honors
- Merit-based Scholarship, awarded to the top 1% of the cohort based on GPA and number of ECTS
- Thesis: "Depth estimation through temporal virtual pattern projection"
- Coursework: Foundations of Computer Science, Math Analysis I-II, Physics, Linear Algebra, Applied Math, Electronics

Liceo Niccolò Copernico Bologna *September 2016 – July 2021*
Scientific High School *Bologna, Italy*

- Final Outcome: 100/100
- Awarded as a National Finalist at the Bocconi Mathematical Games 2017

Work Experience

Flexion Robotics *March 2026 – July 2026*
Intern - Robotics Software Engineer *Zurich, Switzerland*

- Advanced the Python/C++ internal software deployment tool with CI workflows, model-artifact caching, and a network host resolution refactor for portability, driving a 15% increase in deployment speed with improved reliability.
- Led end-to-end design of an AWS-hosted observability platform that collects and analyzes robot performance data to auto-detect regressions, leading to a 40% drop in robot crash rate

Barclays *June 2023 – August 2023*
Summer Intern - Developer Analyst *Glasgow, United Kingdom*

- Developed a FIX-based market order routing platform in Python/C++ using gRPC, ActiveMQ, and QuickFIX frameworks
- Led the internal development of tools previously outsourced, reducing third-party licensing costs by over 85%
- Reduced average order path latency from 1s to 30ms with enhanced system control and security

Selected Projects

OpenICE-SLAM

- A novel neural implicit SLAM (Simultaneous Localization And Mapping) framework that jointly reconstructs geometry, appearance, and open-set semantic labels in an online manner for 3D scene understanding
- Project developed at ETH Computer Vision and Geometry (CVG) lab

Bridging the Gap: Enabling Soft Actor-Critic for High Performance Legged Locomotion rsl-rl-sac

- ETH Semester Research Project at Robotics Systems Lab - Grade: 6.0/6.0
- A Soft Actor-Critic extension to the RSL-RL reinforcement learning training library that diagnoses and fixes the core implementation flaws behind SAC's underperformance, enabling it to match or exceed PPO on locomotion across multiple robot platforms

Skills and Certifications

Coding Languages: Java, C/C++, Python

Technologies and Frameworks: SQL, DB2, OOP, Git, Docker, MATLAB, PyTorch, scikit-learn, NumPy, pandas

Languages: Italian (Native), English (IELTS Academic 7.5 - C1)

Other: First Level Tennis Instructor of the Italian Tennis and Padel Federation (FITP)

Practiced Sports: Tennis | Skiing | Snowboarding | Surfing | Gym